UNIVERSITY OF BIRMINGHAM

School of Computer Science

Data Structures & Algorithms

Sample Exam Questions

Data Structures & Algorithms

1. Consider a recorded embedded B+Tree where each leaf block can contain a maximum of 3 records, each internal block can contain a maximum of 3 keys, all record blocks in the tree are fully occupied with 3 records each and the records have key values: $5, 10, 15, \ldots,$ and there are 4 record blocks in the file

   (i) Draw this tree in a single diagram **[5 marks]**

   (ii) Draw the resulting tree after inserting a record with key value 33 into this tree **[10 marks]**

   (iii) State how many disk block writes are necessary to carry out the insertion above and explain exactly which writes were required. **[5 marks]**

   Topics: 06 - B-Trees

2. Consider inserting following values into a max priority queue (where the highest priority is the highest numeric value) in the given order:

   10, 50, 25, 80, 35, 20, 65, 40, 30, 75, 15, 45, 90

   (i) Assuming a Binary Heap Tree is used to implement the priority queue, draw the resulting binary heap tree both in tree form and in underlying array form using the standard array representation for binary heap trees **[5 marks]**

   (ii) Assuming a Binomial Heap is used to implement the priority queue, draw the resulting Binomial Heap **[10 marks]**

   (iii) Explain the sequence of steps on both implementations to delete the highest priority element and draw the resulting data structures. **[10 marks]**

   Topics: 07 - Priority Queues and Heap Trees

3. Quicksort is being applied to an array containing, in the given order,

   5, 3, 8, 7, 2, 9, 1, 6, 4

   Assume that the pivot selection strategy is to pick the first element in the array, and that the in-place (unstable) partitioning algorithm is used.

   For the first execution of the partition algorithm only, draw the array contents initially and immediately after **each** execution of the call to swap in this algorithm, identifying on your drawing which index location the variables `leftmark` and `rightmark` refer to. **[10 marks]**

   Topics: 08 - Sorting

4. The objects A to G are inserted, in this order, into a hash table. The Hash1 and Hash2 values for these objects are:

	A	B	C	D	E	F	G
Hash1:	17	31	6	14	22	4	5
Hash2:	3	9	5	1	3	7	1

Assume that, in all cases, a maximal load factor of 50% is maintained and that the underlying array has initial capacity of 4 and is doubled in capacity as necessary to maintain the maximal load factor.

(i) Draw a diagram of the resulting hash table if it is implemented with **direct chaining**, where Hash1 is used as the hash value of each object. **[5 marks]**

(ii) Draw a diagram of the resulting hash table if it is implemented with **linear probing**, where Hash1 is used as the hash value of each object. **[5 marks]**

(iii) Draw a diagram of the resulting hash table if it is implemented with **double hashing**, where Hash1 is used as the primary hash value of each object and Hash2 is used as the secondary hash value of each object. **[5 marks]**

Topics: 09 - Hashing and Hash Tables

5. Consider a weighted undirected graph with 5 nodes A, B, C, D, and E, with all weights in the range 1 to 9.

(i) Draw such a graph so that, when the Dijkstra algorithm is applied to it to find the shortest path from A to B, at every iteration of the algorithm a shorter path from A to B is found. **[5 marks]**

(ii) Demonstrate the execution of the algorithm on this graph by writing out the table of execution steps in the same format as was given in the lectures. **[10 marks]**

Topics: 10 - Graphs